PROJECT REPORT

Early Hospital Readmission in Patients with Diabetes: A Descriptive SQL Analysis

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SQL · Descriptive Data Analysis

INTRODUCTION

Diabetes mellitus is a major global public health concern, affecting approximately 589 million adults aged 20–79 years (International Diabetes Federation [IDF], 2025). Unplanned hospital readmissions represent a challenge in the care of patients with diabetes, prompting numerous studies to examine characteristics associated with this outcome (Kukde et al., 2024; Soh et al., 2020). Reported characteristics include comorbidities, age, sex, ethnicity, and various clinical and hospital-related conditions. Polypharmacy has also been documented among older adults with diabetes (Kukde et al., 2024; Satapathy et al., 2025; Soh et al., 2020). Predictive models have additionally been developed to estimate the probability of readmission in this population (Dong et al., 2025; Salim & Ibrahim, 2026). Furthermore, the *Diabetes 130-US Hospitals for Years 1999–2008* dataset has been used in studies examining hospital readmission and HbA1c measurement (Clore et al., 2014; Salim & Ibrahim, 2026; Strack et al., 2014). Analyzing readmission patterns and their distribution across patient and hospitalization characteristics is relevant to understanding this outcome, informing future research on continuity of care, and contributing to the development of preventive strategies aimed at improving patients' quality of life.

OBJECTIVE

To analyze the proportion of patients with recorded diabetes diagnoses in the *Diabetes 130-US Hospitals for Years 1999–2008* dataset (Clore et al., 2014) who experienced a hospital readmission recorded as occurring within less than 30 days, and to describe its distribution according to demographic, clinical, and hospital-related characteristics, recorded diabetes management, and principal diagnoses.

METHODS

Study Design and Data Source

A retrospective, descriptive, observational analysis was conducted using the *Diabetes 130-US Hospitals for Years 1999–2008* dataset (Clore et al., 2014), available from the UCI Machine Learning Repository. The dataset contains 101,766 hospital encounters recorded across 130 US hospitals and integrated healthcare networks between 1999 and 2008. It includes patients with recorded diabetes diagnoses, without restricting the population to patients for whom diabetes was the principal diagnosis or allowing a systematic distinction between type 1 and type 2 diabetes mellitus. Data processing, cleaning, and analysis were performed using SQL queries in DuckDB.

Data Cleaning and Cohort Selection

An analytical cohort was constructed from the original 101,766 hospital encounters using SQL queries. Encounters with discharge dispositions associated with death or hospice care were excluded, and a single eligible encounter was selected for each patient using the minimum

encounter_id as the selection criterion. The number of excluded encounters (2,423) matched that reported by Xiang et al. (2021). However, this agreement concerns the number of exclusions and does not necessarily mean that both studies applied identical procedures. Because exact admission and discharge dates were unavailable, *encounter_id* was used as a proxy for record selection, although it cannot guarantee that the selected record corresponds chronologically to the patient's first encounter. The final analytical cohort consisted of 69,990 patients.

The variables required for the analysis were subsequently derived, including a binary indicator of early hospital readmission. Selected clinical and hospital-related variables were categorized according to the criteria defined for each analysis. The SQL code used for data selection, transformation, and analysis is available in the project repository.

Outcome and Study Variables

The primary outcome was early hospital readmission, defined as a readmission recorded as occurring within less than 30 days, according to the corresponding category of the *readmitted* variable. The proportion of patients with early readmission was calculated relative to the total final cohort.

The variables analyzed included demographic characteristics (age, sex, and race as recorded in the dataset); clinical characteristics (number of medications, number of diagnoses, and principal diagnosis); hospital-related characteristics (length of stay, admission type and source, discharge destination, and medical specialty); and variables related to diabetes management (HbA1c measurement, medication changes, and insulin treatment category). Some variables were grouped into categories to facilitate descriptive comparisons across groups. Principal diagnoses were classified into clinical groups based on their ICD-9 codes.

Descriptive Analysis

A descriptive analysis was conducted using SQL queries to characterize the cohort and examine the distribution of early hospital readmission. Frequencies, percentages, and readmission proportions were calculated for the categories of interest. Differences across groups were also described according to demographic, clinical, hospital-related, and diabetes management characteristics.

To limit the interpretation of estimates based on small groups, minimum sample sizes of 100 records were established for groups defined by sex and race, discharge destination, and treatment; 500 records for hospital care pathways, combining admission type, admission source, and discharge destination; and 200 records for medical specialties. All groups were displayed in the remaining comparisons, but groups with fewer than 100 patients were not interpreted. No hypothesis tests or multivariable models were performed. Accordingly, the findings were interpreted as descriptive patterns, without establishing causal relationships or identifying factors independently associated with readmission.

RESULTS AND DISCUSSION

Cohort Characteristics and Overall Early Readmission Proportion

The final cohort comprised 69,990 patients, of whom 6,285 (8.98%) experienced early hospital readmission, defined as a readmission recorded as occurring within less than 30 days. Strack et al. (2014) analyzed 69,984 records from the same dataset to investigate the relationship between HbA1c measurement and hospital readmission, using a procedure involving the selection of encounters per patient and the exclusion of records associated with death or hospice care. The similarity between the two cohort sizes provides a methodological reference point, although potential differences in data processing and selection criteria should be considered.

Demographic and Clinical Characteristics

By age group, the highest proportions of early readmission were observed among patients aged 80–90 years (10.79%; n = 11,110) and 70–80 years (10.24%; n = 17,751). In the combined analysis of sex and race (as recorded in the dataset), no clear differences in early readmission proportions were observed by sex or race. The proportions across groups ranged from 6.07% to 10.79%, compared with 8.98% for the overall cohort. The largest groups showed similar results: 9.29% among women recorded as Caucasian (n = 26,959) and 9.08% among men recorded as Caucasian (n = 25,346). Estimates for Hispanic and Asian groups should be interpreted cautiously because of their smaller sample sizes (n = 241–807). Approximately 4.4% of the cohort was excluded from this comparison because of missing data or categories not included in the analysis. These findings are descriptive and were not adjusted for age or clinical severity.

In a systematic review and meta-analysis, Soh et al. (2020) identified a higher probability of 30-day readmission among adults aged 65 years or older (OR = 1.23; 95% CI: 1.08–1.40). The review also found a lower pooled probability of readmission among women (OR = 0.94; 95% CI: 0.91–0.97) and White individuals (OR = 0.89; 95% CI: 0.84–0.94), compared with men and non-White individuals, respectively. Thus, the evidence does not consistently support the interpretation that these groups have a higher risk. Regarding readmissions among younger adults, Emons et al. (2016) identified a higher probability of readmission among adults aged 25–44 years in a population hospitalized for hypoglycemia-related episodes. Albrecht et al. (2012), in turn, examined the relationship between severe mental illness and readmission but found no significant association among patients aged 35 years or older. These differences underscore that readmission patterns may vary according to the population studied and the clinical context.

Differences in the proportion of early readmission were also observed according to the number of medications and recorded diagnoses. The highest proportion occurred among patients with 21 or more medications and nine or more diagnoses (11.07%; n = 9,172), followed by those with 21 or more medications and five to eight diagnoses (10.18%; n = 5,410). These findings suggest a higher proportion of readmission among groups with a greater recorded medication and diagnostic burden. Soh et al. (2020) identified associations between 30-day hospital readmission and several clinical characteristics, including certain comorbidities and longer hospital stays. Timple and Kawar (2022), in a study of patients with type 2 diabetes mellitus, found a higher probability of readmission among those with more comorbidities. Similarly, Satapathy et al. (2025) documented a high prevalence of polypharmacy among older adults with diabetes, providing context for the use of multiple medications in this population. However, these studies do not establish that the number of medications or diagnoses recorded in the present analysis constitutes an independent predictor of readmission. These variables may describe aspects of healthcare and recorded clinical complexity,

but they are not, by themselves, direct measures of disease severity, chronic multimorbidity, or sustained polypharmacy.

Regarding HbA1c measurement, 18.35% of patients had a recorded result (n = 12,846), a proportion similar to the 18.4% reported by Strack et al. (2014) using the same dataset. This agreement is descriptively relevant, although it does not constitute independent validation. In the analyzed cohort, the proportion of early readmission was 8.40% among patients with a recorded result, compared with 9.11% among those without one. Strack et al. (2014) found that the association between HbA1c measurement and readmission varied according to the principal diagnosis. Measurement was associated with a lower probability of readmission among patients whose principal diagnosis was diabetes, but not among those with other principal diagnoses. In contrast, Bakeri et al. (2018), in a retrospective study of 5,126 patients with diabetes, found that HbA1c values did not predict 30-day readmission, although higher values were observed among patients with multiple readmissions. These differences highlight that the relationship between HbA1c measurement or values and readmission depends on the population studied and how the outcome is defined.

The early readmission proportion was 9.44% among patients with medication changes (n = 31,497) and 8.60% among those without changes (n = 38,493). However, the latter proportion includes 16,686 patients who received no diabetes medication, whose early readmission proportion was 7.56%. When considering treated patients only, the readmission proportions were approximately 9.4% among those with medication changes (n = 31,497) and among those without them (n = 21,807). Therefore, no meaningful descriptive difference associated with medication changes alone was observed. For insulin treatment categories, the highest proportions were observed among patients whose insulin dose was decreased (10.54%; n = 7,324) or increased (9.84%; n = 6,777). Overall, these patterns are descriptive and do not establish independent associations or causal relationships.

Admission, Discharge, and Principal Diagnosis Characteristics

Readmission proportions also varied according to combinations of admission type, admission source, and discharge destination. The highest value was observed among patients admitted through an emergency admission from an emergency room and discharged to a rehabilitation facility (29.84%; n = 667). When records were grouped by discharge destination, the highest proportions were observed among patients discharged or transferred to rehabilitation facilities (26.36%; n = 1,411) and other inpatient care institutions (20.59%; n = 913). In contrast, the proportion was 6.95% among patients discharged home (n = 44,322) and 9.54% among those discharged home with home health services (n = 8,291). These findings describe differences across recorded discharge destinations without establishing that discharge destination causes subsequent readmission.

In the principal diagnosis analysis, diseases of the circulatory system were the most frequent group, accounting for 30.56% of the cohort and 32.94% of early readmissions. The early readmission proportion within this group was 9.68%. Relatively high proportions were also observed within the injury and poisoning group (10.80%) and the grouped V/E codes (13.49%), although the latter included 919 patients. Diabetes as the principal diagnosis accounted for 8.21% of the cohort.

The relevance of cardiovascular diseases in the context of hospital readmission has also been described in the literature. Hansen et al. (2022) documented frequent readmissions among patients with diabetes undergoing percutaneous coronary intervention, while Fatuyi et al. (2022) reported a 30-day readmission proportion of 17.5% among patients with type 2 diabetes, chronic systolic heart failure, and atrial fibrillation. Ostling et al. (2017) found higher readmission rates among patients with diabetes than among those without diabetes and observed differences in readmission causes

depending on whether diabetes was the principal or a secondary diagnosis. Collectively, these studies provide context for the importance of cardiovascular diseases and diabetes in hospital readmissions. However, their results are not directly comparable with those of the present analysis because of differences in populations and selection criteria. The interpretation of diagnostic categories depends on the classification used; therefore, heterogeneous groups should be considered descriptive and not equivalent to individual clinical diagnoses.

Length of Hospital Stay

Length of hospital stay showed a descriptive pattern in which the proportion of readmission recorded within less than 30 days increased as hospital stays became longer. The proportion was 7.53% among patients with stays of 1–3 days (n = 35,146), 9.91% among those with stays of 4–7 days (n = 25,075), and 11.79% among patients with stays of eight days or longer (n = 9,769). The average hospital stay was also 4.79 days among patients with early readmission, 4.42 days among those readmitted after more than 30 days, and 4.12 days among those without a recorded readmission. Overall, these findings show a higher proportion of early readmission among groups with longer hospital stays.

This pattern is consistent with the findings reported by Rachoin et al. (2020), who examined the relationship between length of hospital stay and readmission. However, the present results are descriptive and cannot determine whether length of stay is independently associated with early readmission or establish a causal relationship.

Early Hospital Readmission by Medical Specialty

Among medical specialties with at least 200 records, oncology had the highest proportion of early readmission (18.05%; n = 205), followed by vascular surgery (11.98%; n = 359) and psychiatry (11.09%; n = 613). In contrast, the lowest proportions were observed in obstetrics and gynecology (3.71%; n = 593) and neurosurgery (5.20%; n = 404). However, medical specialty was not recorded for 33,654 patients (48.1% of the cohort), limiting the interpretation and generalizability of these comparisons. The observed differences should be considered exploratory because no adjustments for potential confounding variables or statistical significance tests were performed.

Limitations

This analysis uses a retrospective dataset of hospital encounters from 130 hospitals in the United States between 1999 and 2008; therefore, its findings may not be generalizable to other healthcare systems or current clinical practice. The analyzed population is not restricted exclusively to patients with type 1 or type 2 diabetes mellitus but includes hospitalization records associated with diabetes diagnoses. Furthermore, the absence of exact admission and discharge dates prevents the precise reconstruction of hospitalization chronology. Consequently, selecting the encounter with the minimum identifier does not necessarily guarantee that it corresponds to the patient's first chronological encounter.

Additional limitations include missing data, age recorded in ten-year intervals, categorical HbA1c information, and limited information about treatment continuity after discharge. Moreover, the uneven availability of certain variables, such as medical specialty, restricts comparisons across groups. Finally, the analysis is descriptive and does not include adjustments for potential confounding variables or statistical significance tests. Therefore, it cannot establish causal relationships, identify independent associations, or determine whether all readmissions are directly related to diabetes.

CONCLUSIONS

In the cohort of 69,990 patients analyzed, the proportion of early hospital readmission, defined as a readmission recorded as occurring within less than 30 days, was 8.98%. Descriptive differences were observed according to demographic, clinical, and hospital-related characteristics, recorded diabetes management, principal diagnoses, and medical specialty. Among the most notable patterns, readmission proportions were higher in groups with longer hospital stays and a greater recorded medication and diagnostic burden. Differences were also observed according to discharge destination and the recording of certain clinical and treatment-related variables.

Overall, these findings characterize the distribution of early readmission in the population studied and provide context in relation to previous literature. Nevertheless, they should be interpreted in light of the dataset's limitations and the descriptive study design. The results do not establish causal relationships or identify independent risk factors, but they provide a basis for future research using adjusted statistical analyses to explore whether the observed patterns persist when multiple clinical and hospital-related characteristics are considered simultaneously.

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